Functional data analysis

5. Mathematical framework for functional data

Alfredas Račkauskas, Jurgita Markevičiūtė

Lecture notes, 2026

Objective are:

- *to provide a basic introduction to function spaces,*
- *to discuss how functional objects can be modelled using this framework.*

Outline of the lecture

Sample space and random sample

Sample space $\mathbb{S}$ is a set where statistical data come from.

- Each element of a sample space is called a sample point.

A sample of size n taken from $\mathbb{S}$ is a collection of n sample points

$$x_1, x_2, \ldots, x_n \in \mathbb{S}.$$

- If $\mathbb{S} \subseteq \mathbb{R}$ we have a univariate sample.
- If $\mathbb{S} \subseteq \mathbb{R}^k$ we have a k-variate (multivariate) sample.
- If $\mathbb{S} \subseteq \mathbb{R}^{[0,1]}$ - set of functions defined on $[0, 1]$, we have a functional sample.

- Mathematical statistics interprets the sample $x_1, \ldots, x_n$ taken from $\mathbb{S}$ as realization of n random points $X_1, \ldots, X_n$ in $\mathbb{S}$ (points in $\mathbb{S}$ depending on an outcome of an underlying random experiment).
- In order to perform statistical inference from functional sample, e.g. hypothesis tests and point estimation with associated measures of uncertainty, we have to treat the observed random functions as random elements of some space.
- This means that we have to fix a functional framework.
- One way to do this is to interpret functional sample as trajectories of random processes with certain paths properties.

Regularity properties of stochastic processes

- How can we deal with *regularity properties* such as continuity, differentiability, or Lipschitz *for the sample functions* of the stochastic process $X = (X(t), t \in [0, 1])$?
- The measure theoretic difficulty arises from the fact that the property that sample function be continuous on $[0, 1]$, or other similar requirements, cannot be expressed in terms of innumerably many points $t \in [0, 1]$.
- So we cannot immediately assert that the event is measurable.

- A simple, but not always sufficient, way of dealing with the continuity of stochastic process is by concept of continuity in probability.

Definition

A stochastic process $X = (X(t), t \in [0,1])$ is continuous in probability at t if $X(t+h)$ tends to $X(t)$ in probabilioty as $h \to 0$, so that for every $\varepsilon > 0$

$$\lim_{h \to 0} \mathrm{P}(|X(t+h) - X(t)| > \varepsilon) = 0.$$

If X is continuous in probability for every t, we call X just continuous in probability.

- Continuity in probability is an important concept, but it is not to be confused with continuity of sample functions.

- To be able to resolve regularity questions, we use the concept of *modification (versions) of stochastic process*.

Definition

For two processes $X = (X(t), t \in [0,1])$ and $Y = (Y(t), t \in [0,1])$, we say they are modifications of each other if

$$P(X(t) = Y(t)) = 1$$

for all $t \in [0,1]$.

- If Y is a modification of X then $X \overset{\mathcal{D}}{\sim} Y$. Indeed, for any $t_1, \ldots, t_d \in [0,1]$ we have

$$\begin{aligned}
1 &\geqslant P(X(t_1) = Y(t_1), \ldots, X(t_d) = Y(t_d)) \\
&= 1 - P(X(t_1) \neq Y(t_1) \cup \cdots \cup X(t_d) \neq Y(t_d)) \\
&\geqslant 1 - \sum_{i=1}^{d} P(X(t_i) \neq Y(t_i)) \\
&= 1.
\end{aligned}$$

- The stochastic process can have modifications with quit different but needed sample properties.

Example

Consider the probability space $(\Omega, \mathcal{F}, \mathsf{P})$, where $\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}_{[0,1]}$ and P - the uniform probability ($\mathsf{P}([a,b]) = b - a$). Define the processes $X = (X(t), t \in [0,1])$ and $Y = (Y(t), t \in [0,1])$ by

$$Y(t,\omega) = 0, \quad X(t,\omega) = \begin{cases} 0, & \text{if} \quad t \neq \omega, \\ 1 & \text{if} \quad t = \omega. \end{cases}$$

Then Y is a modification of X and have different samples.

Examples

Consider a stochastic process $X = (X(t), t \in [0,1])$.

Theorem

If there are finite, positive constants β, α, C such that

$$\mathrm{E}|X(t) - X(t+h)|^{\alpha} \leq C|h|^{1+\beta} \tag{KC}$$

for all $t, t+h \in [0,1]$, X has a modification $\widetilde{X}$ for which sample functions are continuous on $[0,1]$.

- As an example consider a standard Wiener process $W = (W(t), t \in [0,1])$.
- We have

$$\mathrm{E}|W(t+h) - W(t)|^{\alpha} = \frac{1}{\sqrt{2\pi|h|}} \int_{-\infty}^{\infty} |u|^{\alpha} e^{-u^2/2|h|} du = C|h|^{\alpha/2}$$

  and taking $\alpha > 2$ we see that the condition (KC) is satisfied.
- This justifies the requirement of paths continuity in the definition of Wiener process.

Theorem

If there are finite, positive constants p, β, C such that

$$\mathrm{E}|X(t+h)+X(t-h)-2X(t)|^p \leq C|h|^{1+p+\beta} \qquad \text{(KC1)}$$

for all $t, t+h \in [0,1]$, X has a modification $\widetilde{X}$ for which sample functions are continuously differentiable on $[0,1]$.

- For a standard Wiener process we have

$$\mathrm{E}|W(t+h)+W(t-h)-2W(t)|^p = C|h|^{p/2}$$

  and we see that the condition (KC1) is not satisfied.
- Actually it is known that the paths of a Wiener process are nowhere differentiable with probability one.

Linear function spaces

- To simplify the notation, we assume that all functions are defined on an interval $\mathcal{I}$ (if not fixed the interval $\mathcal{I}$ can be finite or infinite).
- If x is such a function, we shall write $x = (x(t), t \in \mathcal{I})$ (equivalent notation include $x : \mathcal{I} \to \mathbb{R}$; $x(t), t \in \mathcal{I}$), whenever we need to stress the interval $\mathcal{I}$.
- Multiplication by real numbers and addition of functions are defined point-wise, i.e.

$$(ax + by)(t) = ax(t) + by(t), \quad t \in \mathcal{I}.$$

Definition

A set $\mathbb{S} \subset \mathbb{R}^{\mathcal{I}}$ of functions defined on the interval $\mathcal{I}$ forms a ***linear function space*** if

$$ax + by \in \mathbb{S}$$

for all $x, y \in \mathbb{S}$ and $a, b \in \mathbb{R}$.

- In particular, beginning with a subset A of a linear space $\mathbb{S}$, we can create a *subspace* (which itself is a linear space) by forming a new set that contains all the finite dimensional linear combinations of its elements.
- This is referred to as the *linear span* of A and denoted by $\mathrm{span}(A)$:

$$\mathrm{span}(A) = \Big\{ \sum_{k=1}^{n} \lambda_k x_k : \quad x_k \in A, \lambda_k \in \mathbb{R}, n \in \mathbb{N} \Big\}.$$

- Hence,

$$\mathrm{span}\{x\} = \{\lambda x : \quad \lambda \in \mathbb{R}\};$$

and

$$\mathrm{span}\{x_1, x_2\} = \{\alpha x_1 + \beta x_2 : \ \alpha, \beta \in \mathbb{R}\}.$$

- The linear span concept can be sharpened somewhat by introducing the idea of a basis.

Definition (of linear independence)

Let $\mathbb{S}$ be a linear function space.

- Functions $v_1, \ldots, v_k \in \mathbb{S}$ is said to be *linearly independent* if $\sum_{i=1}^{k} a_i v_i = 0$ entails that $a_i = 0$ for all i.
- Functions $v_1, v_2, \ldots$ are *linearly independent* if for each $k > 1$, functions $v_1, \ldots, v_k$ are linearly independent.

- A linear space $\mathbb{S}$ is finite dimensional and, moreover, d-dimensional (denoted $\dim(\mathbb{S}) = d$) if $\mathbb{S}$ contains d linearly independent elements and if any $d+1$ elements of $\mathbb{E}$ are linearly dependent.

- It is necessarily the case that if $\{v_1, \ldots, v_d\} \subset \mathbb{S}$ is a basis and v is any element of $\mathbb{S}$ there are coefficients $b_1, \ldots, b_d$ such that

$$v = \sum_{j=1}^{d} b_j v_j.$$

- If a linear space $\mathbb{S}$ has arbitrarily many linearly independent elements, then it is infinite dimensional (denoted $\dim(\mathbb{S}) = \infty$).

Example

Elementary polynomials $\{t^0, t, t^2, t^3, \ldots\}$ are independent functions on any non-empty interval. Hence, $\dim(\mathbb{R}^{\mathcal{I}}) = \infty$.

Normed function spaces

Definition

A linear function space $\mathbb{S}$ is ***normed*** if for each $x \in \mathbb{S}$ there is defined a norm $||x|| \in \mathbb{R}$ such that

(*i*) $||x|| \geqslant 0$, for all $x \in \mathbb{S}$,

(*ii*) $||x|| = 0$ if and only if $x = 0$,

(*iii*) $||ax|| = |a| \cdot ||x||$, for all $x \in \mathbb{S}$ and $a \in \mathbb{R}$,

(*iv*) $||x + y|| \leqslant ||x|| + ||y||$, for all $x, y \in \mathbb{S}$ (triangle inequality).

- The norm allows to define limits of sequences of functions.

Definition [of convergence]

A sequence (x_n) in the normed function space $\mathbb{S}$ ***converges*** to $x \in \mathbb{S}$ if

$$\lim_{n\to\infty} ||x_n - x|| = 0.$$

- We denote $x_n \to x$ (as $n \to \infty$) or $\lim_{n\to\infty} x_n = x$.

We say that (x_n) in $\mathbb{S}$ is a ***convergent sequence*** if there is an $x \in \mathbb{S}$ such that $\lim_{n\to\infty} x_n = x$.

Definition [of Banach space]

A normed linear function space $\mathbb{S}$ is ***Banach space*** provided the following criterion of convergence holds: a sequence (x_n) in $\mathbb{S}$ is convergent if and only if

$$\lim_{n,m\to\infty} ||x_n - x_m|| = 0.$$

Basis in Banach spaces

- As we have already seen, basis expansions play an important role in methodology for functional data.
- Let $\mathbb{S}$ be a normed linear function space. We say that a set of functions $\{\Lambda_1, \Lambda_2, \dots\} \subset \mathbb{S}$ is a basis in $\mathbb{S}$ if every $x \in \mathbb{S}$ admits a unique expansion

$$x = \sum_{j=1}^{\infty} a_j \Lambda_j,$$

that is

$$\lim_{N \to \infty} \left\| x - \sum_{j=1}^{N} a_j \Lambda_j \right\| = 0.$$

The space of continuous functions I

- We consider functions defined on the interval $[a, b]$.
- Recall the function f is continuous at the point $t_0 \in [a, b]$ if for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|t - t_0| < \delta \ \text{ yields } \ |f(t) - f(t_0)| < \varepsilon.$$

- The function f is continuous if it is continuous at every point of $[a, b]$.
- We denote by $C[a, b]$ the set of all continuous functions defined on $[a, b]$. Elements of $C[a, b]$ we shall denote by $f, g, \ldots$

The set $C[a, b]$ is a *linear space of functions*.

- Indeed, we know from analysis that if f, g are continuous functions then such are $f + g$ and af for any $a \in \mathbb{R}$.

The space of continuous functions II

- For each function $f \in C[a,b]$ we associate a norm $||f||$ defined by
$$||f|| = \max_{a \leq t \leq b} |f(t)|.$$
- Convergence in $C[a,b]$ coincides with uniform convergence of functions.
- The distance between two functions in $C[a,b]$ is then the norm of its difference, i.e.,
$$d(f,g) = ||f - g|| = \max_{a \leq t \leq b} |f(t) - g(t)|.$$
- Two functions are close (similar) with respect to this distance if their values are close at every point of $[a,b]$.

The space of continuous functions III

Exercise

Check, that $||f||$ is a norm.

Theorem

The space $C[a, b]$ endowed with the norm $||f||$ is a Banach space.

Faber-Schauder basis in $C[0,1]$

- Set $\mathcal{D}_j = \{(2k-1)2^{-j},\ 1 \leqslant k \leqslant 2^{j-1}\}$, $j \geqslant 1$.
- Set also $\mathcal{D}_0 = \{0, 1\}$ and

$$\mathcal{D} = \bigcup_{j=0}^{\infty} \mathcal{D}_j.$$

  Then $\mathcal{D}$ is called the *set of dyadic numbers*.
- $r \in \mathcal{D}$ is said to be of level j if $r \in \mathrm{D}_j$.
- For $r \in \mathrm{D}_j$ we denote

$$r^- :\ = r - 2^{-j}, \qquad r^+ :\ = r + 2^{-j}.$$

- We have

$$r^+, r^- \in \mathcal{D}_0 \cup \mathrm{D}_1 \cup \cdots \cup \mathcal{D}_{j-1}, \quad \text{if} \ \ r \in \mathcal{D}_j.$$

- The intervals $[r^-, r]$, $(r, r^+]$, for $r \in \mathcal{D}_j$ are called dyadic of the level j.

- For each $r \in \mathcal{D}_j$, $j \geqslant 1$, the function Λ_r is defined by:

$$\Lambda_r(t) = \begin{cases} 2^j(t - r^-), & \text{if } t \in (r^-, r]; \\ 2^j(r^+ - t), & \text{if } t \in (r, r^+]; \\ 0 & \text{otherwise.} \end{cases}$$

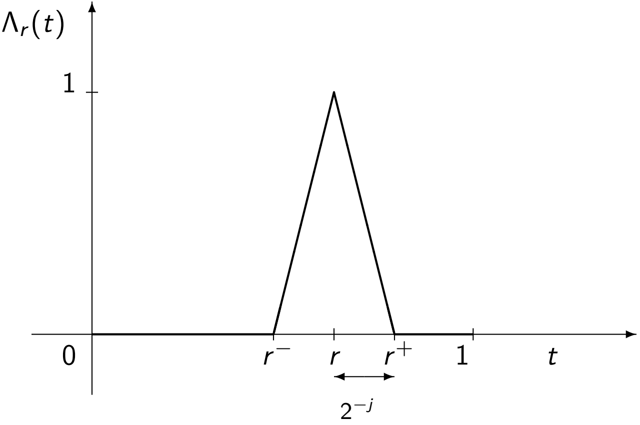
  $t \in [0, 1]$.
- Moreover we set

$$\Lambda_0(t) = 1 - t, \qquad \Lambda_1(t) = t, \quad t \in [0, 1].$$

- The set

$$\{\Lambda_r, r \in \mathrm{D}\}$$

  is called Faber–Schauder system of functions.



Faber–Schauder functions Λ_r

Theorem

Faber–Schauder system $\{\Lambda_r, r \in \mathrm{D}\}$ is an basis in $C[0,1]$, that is, every $f \in C[0,1]$ admits an expansion

$$f(t) = \sum_{r \in \mathcal{D}} \lambda_r(f)\Lambda_r(t) = \sum_{j=0}^{\infty} \sum_{r \in D_j} \lambda_r(f)\Lambda_r(t), \quad t \in [0,1],$$

where the series converge uniformly and

$$\lambda_r(f) = f(r) - \frac{1}{2}(f(r^+) + f(r^-)), \quad r \in D_j, j \geq 1,$$

$\lambda_0(f) = f(0), \quad \lambda_1(f) = f(1) - f(0).$

Hence, if f is continuous,

$$\lim_{J \to \infty} \sup_{0 \leq t \leq 1} \left| f(t) - \sum_{j=0}^{J} \sum_{r \in D_j} \lambda_r(f)\Lambda_r(t) \right| = 0.$$

Random functions in $C[0,1]$

Definition

A function $X : \Omega \to \mathrm{C}[0,1]$ is said to be a random element in $C[0,1]$ (or $C[0,1]$-valued random function) if

$$X(\omega) \in C[0,1] \ \text{ for each } \ \omega \in \Omega.$$

- Any random process $X = (X(t), t \in [0,1])$ that satisfies the condition (KC) can be treated as $C[0,1]$-valued random function by considering a corresponding version.

- By definition a Wiener process is a $C[0,1]$-valued random function.
- The following Theorem provides the expansion of a standard Wiener as a series of triangular functions.

Theorem

Let $\{\xi_r, r \in \mathcal{D}\}$ be a sequence of independent $\mathcal{N}(0,1)$ random variables. Then

$$W(t) := \xi_1 \Lambda_1(t) + \sum_{j=1}^{\infty} \sum_{r \in \mathcal{D}_j} 2^{-(j+1)/2} \xi_r \Lambda_r(t), \quad t \in [0,1],$$

is a standard Wiener process with continuous sample paths. Removing the term $\xi_1 \Lambda_1$ gives a Brownian bridge.

Inner product spaces

Definition

A linear function space $\mathbb{S}$ forms an inner product space if for any two $f, g \in \mathbb{S}$ there is defined an inner product $\langle f, g\rangle$ so that

(*i*) $\langle f, f\rangle \geq 0$ and is zero only if $f = 0$,

(*ii*) $\langle f, g\rangle = \langle g, f\rangle$,

(*iii*) for any real numbers a, b, and any $f, g, h \in \mathbb{S}$,

$$\langle ag + bh, f\rangle = a\langle g, f\rangle + b\langle h, f\rangle$$

- The inner product allows us to entertain the notion of orthogonality, which makes the geometric structure of inner product space very similar to that of the usual finite dimensional Euclidean space, at least at an intuitive level.

Definition [of orthogonality]

- We say that functions f and g in the inner product space $\mathbb{S}$ are orthogonal, denoting $f \perp g$, if $\langle f, g \rangle = 0$.
- We say that the sets of functions $A, B \subset \mathbb{S}$ are orthogonal, denoting $A \perp B$, if $f \perp g$ for all $f \in A$ and $g \in B$.

- The inner product also allows us to introduce the *associated norm*

$$||f|| = \sqrt{\langle f, f\rangle}, \quad f \in \mathbb{S}.$$

- There is a very useful relation between the inner product of two functions and their norms known as the Cauchy-Schwarz inequality:

$$|\langle f, g\rangle| \leq ||f|| \cdot ||g||.$$

Definition

Infinite dimensional inner product space is called a *Hilbert space* if it is a Banach space with respect to the associated norm.

Orthonormal basis

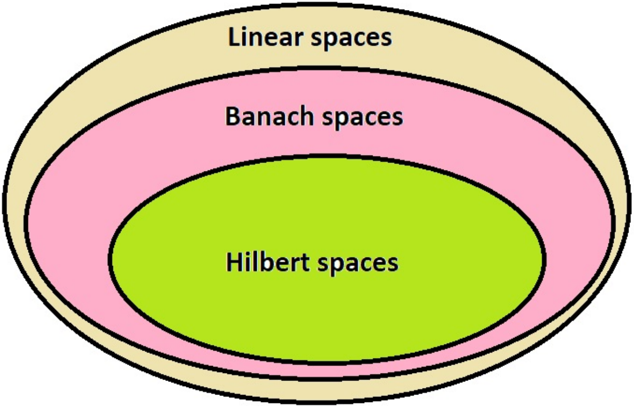
- Let $\mathcal{H}$ be a Hilbert function space.
- We say that a set of functions $\{e_1, e_2, \dots\}$ is a basis in $\mathcal{H}$ if every $x \in \mathcal{H}$ admits a unique expansion

$$x = \sum_{j=1}^{\infty} a_j e_j,$$

  meaning that $\lim_{N\to\infty} \left\| x - \sum_{j=1}^{N} a_j e_j \right\| = 0$.
- We say that $\{e_1, e_2, \dots\}$ is an *orthonormal basis*, if, in addition, $\langle e_j, e_k \rangle = 0$ whenever $j \neq k$ and $||e_j|| = 1$.
- For an orthonormal basis, $a_j = \langle x, e_j \rangle$, and we have the following Parseval's equality:

$$||x||^2 = \sum_{k=1}^{\infty} \langle x, e_k \rangle^2.$$

- If there is an orthonormal basis in $\mathcal{H}$ then the Hilbert space $\mathcal{H}$ is said to be *separable*.


Linear spaces
Banach spaces
Hilbert spaces

Square integrable functions I

- We consider functions that are defined on the interval $[0, 1]$ for simplicity.
- If the argument t of a function f is in the interval $[a, b]$, then we simply consider the function f^* defined by $f^*(u) = f(t)$, where $u = (t - a)/(b - a)$.

Definition

A function f is said to be square integrable if[a]

$$\int_0^1 f^2(t)\,\mathrm{d}t < \infty.$$

We denote the set of all square integrable functions by L_2 (or $L_2([0, 1])$ if there is a need to stress the interval).

[a] We consider Lebesgue integrals.

Square integrable functions II

- Students aware of measure theoretic foundations of integration, will recall that the elements of L_2 and operations on them do not need to be defined for every point t, they are defined for almost all points t.
- We will ignore measure-theoretic considerations, but just we agree that two functions $f, g \in L_2$ are equal, denoting $f = g$, provided
$$\int_0^1 (f(t) - g(t))^2 \,\mathrm{d}t = 0.$$
- Note that $C[0,1] \subset L_2([0,1])$ in a sense that each continuous function on $[0,1]$ being bounded is square integrable, i.e. if $f \in C[0,1]$ then $\int_0^1 f^2(t)\,\mathrm{d}t < \infty$, hence $f \in L_2([0,1])$, but the roles of f being in $C[0,1]$ and that of f being in L_2 are different.

Square integrable functions III

  - $(f(t), t \in [0,1])$ as continuous function is a concrete object whereas the L_2 space element f is an abstract representation of f; for instance, for any t, the symbol $f(t)$ has no meaning as elements of L_2 are equivalence classes of functions.

- Square integrable functions form a vector space: if $f, g \in L_2$, then $af + bg \in L_2$ for any real numbers a and b. Indeed, applying the inequality

$$(a+b)^2 \leq 2(a^2+b^2), \quad a, b \in \mathbb{R},$$

  we deduce

$$\int_0^1 (af(t) + bg(t))^2 \,\mathrm{d}t \leq 2a^2 \int_0^1 f^2(t)dt + 2b^2 \int_0^1 g^2(t)dt.$$

- What makes the space L_2 so convenient is that we can define the *inner product* of two functions f and g by

$$\langle f, g \rangle := \int_0^1 f(t)g(t)dt.$$

- This inner product has the properties that we need for an *inner product*:
  1. $\langle f, f \rangle \geq 0$ and is zero only if $f = 0$,
  2. $\langle f, g \rangle = \langle g, f \rangle$,
  3. for any scalars (real numbers) a and b, and functions f, g and h,

$$\langle \alpha g + \beta h, f \rangle = \alpha \langle g, f \rangle + \beta \langle h, f \rangle.$$

- The *associated norm* is defined by

$$||f|| = \sqrt{\langle f, f\rangle} = \Big(\int_0^1 f^2(t)dt\Big)^{1/2}.$$

- The Cauchy-Schwarz inequality then reads as:

$$\begin{aligned}|\langle f, g\rangle| &= \Big|\int_0^1 f(t)g(t)dt\Big| \\ &\leq \Big(\int_0^1 f^2(t)dt\Big)^{1/2}\Big(\int_0^1 g^2(t)dt\Big)^{1/2} = ||f|| \cdot ||g||.\end{aligned}$$

Theorem

The space L_2 endowed with the norm $||f|| = \sqrt{\langle f, f\rangle}$ is a Banach space. Hence, L_2 is a Hilbert space.

Trigonometric basis

Theorem

Each of the following sets of functions

- $B_1 = \{f_0(t) = 1, f_n(t) = \sqrt{2}\cos(n\pi t), n \geq 1\};$
- $B_2 = \{g_n(t) = \sqrt{2}\sin(n\pi t), n \geq 1\};$
- $B_3 = \{h_0(t) = 1, h_{2n-1}(t) = \sqrt{2}\sin(2n\pi x),$
  $h_{2n}(t) = \sqrt{2}\cos(2n\pi t), n \geq 1, \}$

is orthonormal bases for $L_2[0, 1]$.

The space $L_2(q)$

- Assume q is a density function on $[0, 1]$:

$$q(t) \geq 0 \text{ and } \int_0^1 g(t)dt = 1.$$

- We denote by $L_2(q)$ the set of all square q-integrable functions f:

$$\int_0^1 f^2(t)q(t)dt < \infty.$$

- We agree that two functions $f, g \in L_2(q)$ are equal, denoting $f = g$, provided

$$\int_0^1 (f(t) - g(t))^2 q(t)dt = 0.$$

- Square q-integrable functions form a vector space: if $f, g \in L_2(q)$, then $af + bg \in L_2(q)$ for any real numbers a and b.

- We define the inner product of two functions f and g by

$$\langle f, g \rangle = \langle f, g \rangle_q := \int_0^1 f(t)g(t)q(t)dt$$

- and associated norm by

$$||f|| = ||f||_q := \Big(\int_0^1 f^2(t)q(t)dt \Big)^{1/2}.$$

Theorem

The vector space $L_2(q)$ endowed with the inner product $\langle f, g \rangle_q$ is a Hilbert space.

- The standard approach to creating an orthonormal basis is the Gram-Schmidt algorithm of the orthogonalization of the sequence of elementary polynomials

$$1, t, t^2, \ldots$$

which is described as follows.
- Let $x_n(t) = t^{n-1},\ n = 1, 2, \ldots,\quad t \in [0, 1]$. Define $e_1 = x_1/||x_1||_q$ and $e_i = v_i/||v_i||_q$ for

$$v_i = x_i - \sum_{j=1}^{i-1} \langle x_i, e_j \rangle_q e_j.$$

Then, $\{e_n\}$ is an orthonormal basis for $L_2(q)$.

Random functions in L_2

- Let $(\Omega, \mathcal{F}, \mathrm{P})$ be a probability space.

Definition

A function $X : \Omega \to L_2$ is said to be a *random element* in L_2 (or L_2-valued random function) if $\langle X, f \rangle$ is a random variable for each $f \in L_2$.

- Consider a stochastic process $X = (X(t), t \in [0, 1])$.
- What conditions guarantees that X can be interpreted as random element in L_2?

Theorem

Assume that a stochastic process $X = (X(t), t \in [0, 1])$ has continuous sample functions $(X(t, \omega), t \in [0, 1]$, is continuous function for each $\omega)$. Then X is a random element in L_2.

- In Theorem the notation X is used to denote a stochastic process and a Hilbert space random element.
- It might be useful to emphasize a subtle distinction between the two roles for X.
- The process $(X(t), t \in [0,1])$ is a concrete object that can potentially be observed.
- The Hilbert space element X is an abstract representation of X; for instance, for any t, the symbol $X(t)$ has no meaning as elements of $\mathcal{H}$ are equivalence classes of functions.
- For convenience of notation, we will continue to use X to represent both the process and the Hilbert space element.
- Hence, for any random process $X = (X(t), t \in [0,1]) \in C[0,1]$ we have as well $X \in L_2$.

Linear transformations

- Let $\mathcal{H}$ be a Hilbert space with inner product $\langle x, y\rangle$ and the associated norm $||x|| = \sqrt{\langle x, x\rangle}$ for $x, y \in \mathcal{H}$.

Definition [of linear operator]

A transformation $T : \mathcal{H} \to \mathcal{H}$ is called a *linear operator* if

$$T(af + bg) = aT(f) + bT(g)$$

for any $f, g \in \mathcal{H}$ and any real numbers a, b.

- An operator T is *bounded* if there is a constant M such that for any $x \in \mathcal{H}$, $||Tx|| \le M||x||$.
- The *norm* of a bounded operator is defined by

$$||T|| = \sup_{||x||=1} ||Tx||.$$

- Bounded operators form a vector space, denoted $L(\mathcal{H})$, with the operations defined point-wise, i.e.

$$(T_1 + T_2)(x) = T_1(x) + T_2(x), \quad (aT)(x) = aT(x),$$

  for $x \in \mathcal{H},\ a \in \mathbb{R}$.
- Endowed with the operator norm, the vector space $L(\mathcal{H})$ is a Banach space.

Example

- For $x, y \in \mathcal{H}$ define, so called, tensor product operator $x \otimes y : \mathcal{H} \to \mathcal{H}$ by

$$x \otimes y(z) = \langle x, z \rangle y \quad \text{for} \quad z \in \mathcal{H}.$$

- The operator $x \otimes y$ is linear and bounded. Moreover, its norms is

$$||x \otimes y|| = ||x|| \cdot ||y||.$$

Example

If $x_1, \ldots, x_d, y_1, \ldots, y_d \in \mathcal{H}$, then the operator

$$T := \sum_{i=1}^{d} x_i \otimes y_i$$

is at most d-dimensional as $dim\{Tz, z \in \mathcal{H}\} \leq d$.

Definition [of Hilbert-Schmidt operator]

Assume the space $\mathcal{H}$ is separable. A bounded linear operator $T : \mathcal{H} \to \mathcal{H}$ is said to be Hilbert-Schmidt if for an orthonormal basis $\{e_j, j \geq 1\}$,

$$||T||_{HS}^2 := \sum_{j=1}^{\infty} ||Te_j||^2 < \infty.$$

- The above relation defines the Hilbert-Schmidt norm $||T||_{HS}$.
- We denote by $L_{HS}(\mathcal{H})$ the set of all Hilbert-Schmidt operators on $\mathcal{H}$.

Definition [of eigenfunctions and eigenvalues]

A non-zero function $f \in \mathbb{S}$ is called an *eigenfunction* of T if there exists $\lambda \in \mathbb{R}$ such that

$$Tf = \lambda f.$$

The number λ is called an eigenvalue of T corresponding to f.

- The set $E_\lambda := \{f \in \mathbb{S} : Tf = \lambda f\}$ is a linear subspace of $\mathbb{S}$ and is called the *eigenspace* of T corresponding to the eigenvalue λ.

Definition [of non-negative operator]

An operator $T \in L(\mathcal{H})$ is called non-negative definite if for every $x \in \mathcal{H}$,

$$\langle Tx, x \rangle \geq 0.$$

It is positive definite, if the equality can hold only for $x = 0$.

- Non-negative definite operators are also called: non-negative, positive, positive definite, positive semi-definite.
- Positive definite operators are also called: strictly positive and strictly positive-definite.

Theorem

All eigenvalues of a non-negative definite operator are non-negative. All eigenvalues of a positive-definite operator are positive.

Theorem

Let $T \in L(\mathcal{H})$ be a symetric operator. Then the eigenfunctions f corresponding to distinct eigenvalues are orthogonal.

Proof.

Let ψ_1, ψ_2 be eigenfunctions corresponding to distinct eigenvalues λ, μ, i.e. $\lambda \neq \mu$. Then

$$T\psi_1 = \lambda\psi_1, \quad T\psi_2 = \mu\psi_2$$

and

$$\begin{aligned}
\mu\langle\psi_1, \psi_2\rangle &= \langle\psi_1, \mu\psi_2\rangle = \langle\psi_1, T\psi_2\rangle \\
&= \langle T\psi_1, \psi_2\rangle = \langle\lambda\psi_1, \psi_2\rangle \\
&= \lambda\langle\psi_1, \psi_2\rangle.
\end{aligned}$$

But $\mu \neq \lambda$ so $\langle\psi_1, \psi_2\rangle = 0$. $\square$

Theorem

Let $\mathcal{H}$ be a separable Hilbert space and let $T \in L(\mathcal{H})$ be a symmetric Hilbert-Schmidt operator. Then, $\mathcal{H}$ has an orthonormal basis (e_i) of eigenvectors of T corresponding to eigenvalues λ_i. In addition, the following holds:

1. The eigenvalues (λ_i) has zero as the only possible point of accumulation.
2. The eigenspaces corresponding to distinct eigenvalues are mutually orthogonal.
3. The eigenspaces corresponding to non-zero eigenvalues are finite-dimensional.

In the case of a non-negative ($\langle Tf, f\rangle \geq 0$ for all $f \in \mathbb{S}$) compact self-adjoint operator, then its eigenvalues are non-negative and we may order them as

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq 0.$$

- Hence, any symmetric Hilbert-Schmidt operator $T : \mathcal{H} \to \mathcal{H}$ generates an orthonormal basis (e_i) in $\mathcal{H}$, so that every $x \in \mathcal{H}$ has an expansion

$$x = \sum_{i=1}^{\infty} \langle x, e_i \rangle e_i$$

  meaning that

$$\lim_{N\to\infty} \left\| x - \sum_{i=1}^{N} \langle x, e_i \rangle e_i \right\| = 0.$$

- Moreover, Tx has an expansion

$$Tx = \sum_{i=1}^{\infty} \lambda_i \langle x, e_i \rangle e_i$$

  meaning that

$$\lim_{N\to\infty} \left\| Tx - \sum_{i=1}^{N} \lambda_i \langle x, e_i \rangle e_i \right\| = 0.$$

Integral operators and Mercer's Theorem

- A function $K : [0,1] \times [0,1] \to \mathbb{R}$ is square integrable if

$$\int_0^1 \int_0^1 K^2(s,t)\,\mathrm{d}s\,\mathrm{d}t < \infty.$$

- For such a function we define the integral transformation of a function by

$$(\mathcal{K}f)(t) := \int_0^1 K(t,s) f(s)\,\mathrm{d}s, \quad t \in [0,1].$$

- The function K is called a kernel of the transformation $\mathcal{K}$ whereas $\mathcal{K}$ is called an integral operator with kernel K.

- $\mathcal{K}$ maps L_2 to L_2 and is linear and bounded operator. Indeed, linearity is evident and boundedness follows from

$$||\mathcal{K}x||^2 = \int_0^1 (\mathcal{K}x)^2(t)\,\mathrm{d}t = \int_0^1 \Big(\int_0^1 K(t,s)x(s)\,\mathrm{d}s\Big)^2 \mathrm{d}t$$
$$\leq \int_0^1 \Big[\int_0^1 K^2(t,s)\,\mathrm{d}s \int_0^1 x^2(s)\,\mathrm{d}s\Big]\,\mathrm{d}t = C||x||^2,$$

  by Cauchy-Schwarz inequality, where

$$C = \int_0^1 \int_0^1 K^2(t,s)\,\mathrm{d}s\,\mathrm{d}t.$$

- Hence, $\mathcal{K} \in L(L_2)$.

Theorem

An integral operator $\mathcal{K} : L_2 \to L_2$ corresponding to square integrable kernel is Hilbert-Schmidt operator.

Theorem [Mercer's]

Let the continuous kernel K be symmetric and non-negative definite and $\mathcal{K}$ the corresponding integral operator. Then there is an orthonormal basis $(e_i, i \geq 1)$ of L_2 consisting of eigenfunctions of $\mathcal{K}$ such that the corresponding sequence of eigenvalues $(\lambda_i, i \geq 1)$ are non-negative. The eigenfunctions corresponding to non-zero eigenvalues are continuous on $[0, 1]$ and K has the representation

$$K(s,t) = \sum_{j=1}^{\infty} \lambda_j e_j(s) e_j(t),$$

for all s, t, with the sum converging absolutely and uniformly. ,

$$\lim_{N \to \infty} \sup_{s,t \in [0,1]} \left| K(s,t) - \sum_{j=1}^{N} \lambda_j e_j(s) e_j(t) \right| = 0.$$

Example

- An operator that arises in various settings is the $L_2[0,1]$ integral operator that has the kernel

$$K(s,t) = \min\{s,t\},$$

for $s,t \in [0,1]$. Note that K is the covariance function of the standard Brownian motion process on $[0,1]$.

- The operator that corresponds to K is given explicitly by

$$(\mathcal{K}f)(t) = \int_0^1 K(t,s)f(s)\,\mathrm{d}s = \int_0^t sf(s)\,\mathrm{d}s + t\int_t^1 f(s)\,\mathrm{d}s.$$

- Let us now find its eigenvalues and eigenfunctions by solving

$$\int_0^1 \min\{s, t\} e(s)\,\mathrm{d}s = \lambda e(t)$$

  for λ and e.
- In this regard, first note that this equation is the same as

$$\int_0^t s e(s)\,\mathrm{d}s + t \int_t^1 e(s)\,\mathrm{d}s = \lambda e(t).$$

  Differentiating both sides of this relation produces

$$te(t) + \int_t^1 e(s)\,\mathrm{d}s - te(t) = \lambda e'(t),$$

  i.e.,

$$\int_t^1 e(s) ds = \lambda e'(t).$$

- Differentiating again reveals that

$$e(t) = -\lambda e''(t)$$

for which the general solution is

$$e(s) = a\sin(s/\sqrt{\lambda}) + b\cos(s/\sqrt{\lambda}).$$

- Since $e(0) = 0$, so $b = 0$ in the above.
- Similarly, we have $e'(1) = 0$, so $a\cos(1/\sqrt{\lambda}) = 0$, which leads to

$$1/\sqrt{\lambda} = (2j-1)\pi/2, \quad j = 1, 2, \ldots$$

Thus, the eigenvalues are

$$\lambda_j = \frac{4}{((2j-1)\pi)^2}$$

and the orthonormal eigenfunctions are

$$e_j(t) = \sqrt{2}\sin((2j-1)\pi t/2).$$

Karhunen-Loève expansion

Consider a second order stochastic process $X = (X(t), t \in [0,1])$.

- We say a stochastic process is mean-square continuous if

$$\lim_{h \to 0} \mathsf{E}(X(t+h) - X(t))^2 = 0.$$

- Recall the autocorrelation function $\mathcal{R}_X$ of a stochastic process $X = (X(t), t \in \mathcal{I})$ is given by

$$\mathcal{R}_X(s,t) = \mathsf{E}(X(t)X(s)], \quad t, s \in \mathcal{I}.$$

Theorem

A second order stochastic process $X = (X(t), t \in [0,1])$ is mean-square continuous if and only if its autocorrelation function $\mathcal{R}_X$ is continuous on $[0,1] \times [0,1]$.

Proof. Suppose $\mathcal{R}_X$ is continuous, and note that

$$\begin{aligned} \mathsf{E}(X(t+\varepsilon)-X(t))^2 &= \mathsf{E}(X^2(t+\varepsilon)) - 2\mathsf{E}[X(t+\varepsilon)X(t)] + \mathsf{E}(X^2(t)) \\ &= \mathcal{R}_X(t+\varepsilon, t+\varepsilon) - 2\mathcal{R}_X(t+\varepsilon, t) + \mathcal{R}_X(t,t). \end{aligned}$$

Therefore, since $\mathcal{R}_X$ is continuous,

$$\begin{aligned} \lim_{\varepsilon\to 0} \mathsf{E}(X(t+\varepsilon)-X(t))^2 = \lim_{\varepsilon\to 0} \Big[&\mathcal{R}_X(t+\varepsilon, t+\varepsilon) - 2\mathcal{R}_X(t+\varepsilon, t) \\ &+ \mathcal{R}_X(t,t)\Big] = 0. \end{aligned}$$

That is X is mean-square continuous.

If X is mean-square continuous we proceed as follows

$$
\begin{aligned}
&|\mathcal{R}_X(t+\varepsilon, s+\delta) - \mathcal{R}_X(t,s)| = |\mathrm{E}[X(t+\varepsilon)X(s+\delta)] - \mathrm{E}[X(t)X(s)]| \\
&= \Big|\mathrm{E}[(X(t+\varepsilon) - X(t))(X(s+\delta) - X(s))] + E[(X(t+\varepsilon) - X(t))X(s)] \\
&\qquad + E[(X(s+\delta) - X(s))X(t)]\Big| \\
&\leq \Big|\mathrm{E}[(X(t+\varepsilon) - X(t))(X(s+\delta) - X(s))]\Big| + \Big|E[(X(t+\varepsilon) - X(t))X(s)]\Big| \\
&\qquad + \Big|\mathrm{E}[(X(s+\delta) - X(s))X(t)]\Big| \\
&\leq \Big(\mathrm{E}(X(t+\varepsilon) - X(t))2\Big)^{1/2}\Big(\mathrm{E}((X(s+\delta) - X(s))2\Big)^{1/2} \\
&\qquad + \Big(\mathrm{E}(X(t+\varepsilon) - X(t))^2\Big)^{1/2}\Big(\mathrm{E}X^2(s)\Big)^{1/2} \\
&\qquad + \Big(\mathrm{E}(X(s+\delta) - X(s))^2\Big)^{1/2}\Big(\mathrm{E}X^2(t)\Big)^{1/2},
\end{aligned}
$$

where the last inequality follows from Cauchy-Schwarz inequality. By mean-square continuity of X we have that

$$
\lim_{(\varepsilon,\delta)\to(0,0)} |\mathcal{R}_X(t+\varepsilon, s+\delta) - \mathcal{R}_X(t,s)| = 0.
$$

- Assume that $X = (X(t), t \in [0,1])$ is a centred mean-square continuous stochastic process.
- Then the integral operator $\mathcal{K}_X$ is defined by

$$[\mathcal{K}_X f](t) = \int_0^1 \mathcal{R}_X(t,s) f(s) ds.$$

Lemma

The integral operator $\mathcal{K}_X$ belongs to $L(L_2)$ and the following hold:

1. $\mathcal{K}$ is symmetric
2. $\mathcal{K}$ is non-negative definite.
3. $\mathcal{K}$ is Hilbert-Schmidt.

Proof.

1. This follows trivially from $\mathcal{R}_X(s,t) = \mathcal{R}_X(t,s)$.

2. Since the process X is mean-square continuous, Theorem implies that $\mathcal{R}_X(s,t)$ is continuous. Hence, it is square integrable and as a consequence $\mathcal{K}_X$ is a Hilbert-Schmidt operator.

3. We need to show $\langle \mathcal{K}_X u, u \rangle \geq 0$ for every $u \in L_2$. We have

$$\begin{aligned}
\langle \mathcal{K}_X u, u \rangle &= \int_0^1 \mathcal{K}_X u(s) u(s) ds = \int_0^1 \Big(\int_0^1 \mathcal{R}_X(s,t) u(t) dt \Big) u(s) ds \\
&= \int_0^1 \Big(\int_0^1 \mathrm{E}[X(s)X(t)] u(t) dt \Big) u(s) ds \\
&= \mathrm{E} \int_0^1 \int_0^1 X(s)X(t)u(t)u(s) dt ds \\
&= \mathrm{E} \Big(\int_0^1 X(s)u(s) ds \Big)^2 \geq 0,
\end{aligned}$$

where we used Fubini's Theorem to interchange integrals.

$\square$

- Now, these properties of $\mathcal{K}_X$ allows us to invoke the spectral theorem for symmetric non-negative Hilbert-Schmidt operators to conclude that $\mathcal{K}_X$ has a complete set of eigenvectors (e_i) in L_2 and non-negative eigenvalues (λ_i):

$$\mathcal{K}_X e_i = \lambda_i e_i.$$

- If the zero mean stochastic process X which we fixed in the beginning is assumed to be a random element in L_2, we may use the basis (e_i) of L_2 to expand X as follows,

$$X(\omega) = \sum_i \langle X(\omega), e_i \rangle e_i$$

where the convergence is in L_2:

$$\lim_{N \to \infty} \left\| X(\omega) - \sum_{i=1}^{N} \langle X(\omega), e_i \rangle e_i \right\| = 0.$$

- The following lemma summarizes the properties of the coefficients of the expansion.

Lemma

The coefficients $\langle X, e_i \rangle$ satisfy the following:

1. $\mathsf{E}[\langle X, e_i \rangle] = 0$.
2. $\mathsf{E}[\langle X, e_i \rangle \langle X, e_j \rangle] = \delta_{ij} \lambda_j$.
3. $\mathrm{var}[\langle X, e_i \rangle] = \lambda_i$.

Theorem (Karhunen-Loeve)

Let $X = (X(t), t \in [0,1])$ be a centred mean-square continuous stochastic process with $X \in L_2$. There exist an orthonormal basis (e_i) of L_2 such that for all $t \in [0,1]$,

$$X(t) = \sum_{i=1}^{\infty} \xi_i e_i(t) \;\; \text{in} \;\; L_2,$$

where coefficients ξ_i are given by

$$\xi_i(\omega) = \int_a^b X(t)(\omega) e_i(t) dt$$

and satisfy the following.

1. $\mathsf{E}[\xi_i] = 0$.
2. $\mathsf{E}[\xi_i \xi_j] = \delta_{ij} \lambda_j$.
3. $\mathrm{var}[\xi_i] = \lambda_i$.

Remark

Suppose $\lambda_k = 0$ for some k, and consider the coefficient ξ_k in the expansion. Then, we have by the above Theorem $\mathsf{E}[\xi_k] = 0$ and $\mathrm{var}[\xi_k] = \lambda_k = 0$, and therefore, $\xi_k = 0$ a.s. That is, the coefficient ξ_k corresponding to a zero eigenvalue is zero. Therefore, only ξ_i corresponding to positive eigenvalues λ_i appear in KL expansion of a square integrable, centred, and mean-square continuous stochastic process.

- In the view of the above remark, we can normalize the coefficients ξ_i in a KL expansion and define $\widetilde{\xi}_i = \xi_i/\sqrt{\lambda_i}$. This leads to the following, more familiar, version of KL Theorem.

Corollary

Let $X = (X(t), t \in [a, b])$ be a centred mean-square continuous stochastic process with $X \in L_2([a, b])$. There exist a basis (e_i) of L_2 such that for all $t \in [a, b]$,

$$X(t, \omega) = \sum_{i=1}^{\infty} \widetilde{\xi}_i(\omega) e_i(t) \ \text{ in } \ L_2([a, b]),$$

where $\widetilde{\xi}_i$ are centred mutually uncorrelated random variables with unit variance and are given by,

$$\widetilde{\xi}_i(\omega) = \frac{1}{\sqrt{\lambda_i}} \int_a^b X(t, \omega) e_i(t) dt.$$